

Exploring the efficient computation of permanents of graph Laplacians



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Abstract

In this project, we explore some combinatorial approaches for efficiently approximating the permanents of graph Laplacians. We present a survey of some classical structural results on the permanents of special classes of matrices. We also explore potential recursive expressions for the permanent of the Laplacian for some simple classes of graphs.

1 Introduction

The permanent function is very similar to the determinant function. Given any $n \times n$ matrix $X_{n \times n}$ over complex numbers and the set S_n , which consists of all permutations on n - elements, the permanent of $X_{n \times n}$ is defined as follows

$$\text{per}(X) = \sum_{\sigma \in S_n} \prod_{i=1}^n x_{i, \sigma(i)}$$

where x_{ij} denote $(i, j)^{th}$ entry of the matrix X . Unlike the case of determinant, we do not include the sign of the permutations in permanent.

Throughout the history of matrices, permanents occur in various fields during computations like as computational physics, arithmetic circuits, combinatorics. In quantum computing, there is

an important step based on a sampling called as "boson sampling". The output probabilities of a boson sampling process are given by permanent of sub-matrices of a unitary matrix that describes the underlying system [LM22]. Also in the field of algebraic complexity theory, figuring out the optimal size of arithmetic circuits that computes the permanent is a central problem [GKKS16].

Thus one of the very challenging problem is to calculate the permanent of a matrix efficiently. In [Val79], Valiant showed that computation of permanent of matrices $A \in \{0, 1\}^{n^2}$ is #P complete. Later Jerrum, Sinclair, Vigoda gave the answer to problem restricting to the non-negative matrices in [JSV04, Theorem 1.1] which gives us the existence of a fully polynomial randomized approximation scheme for the permanent of an arbitrary $n \times n$ matrix A with non-negative entries. Now the problem is whether we can say similar thing about permanent of a matrix with arbitrary entries or not. An independent approach by Aaronson in [Aar11] showed that computing even the *sign* of the permanent of a matrix with integer entries specified in binary is #P-hard. Grier and Schaeffer [GS16] showed that even calculating permanent of a real orthogonal matrix and a positive-definite matrix in $\mathbb{Z}^{n \times n}$ is #P-hard. This implies the larger class of positive semi-definite matrices is #P-hard. Quite recently in [AGGS17], it is shown that there is a deterministic polynomial time c^n approximation algorithm for the permanent of positive semi-definite matrices where $c = e^{\gamma+1} \approx 4.84$, where γ is Euler's constant.

We explored to get lower bounds on permanent for diagonally dominant matrix. J. L. Brenner [Bre54] gave upper and lower bound for determinant of diagonally dominant matrices based on the elementary row operation. Since permanent of matrix does not preserve under all elementary row operations, thus it cannot be extended to permanent. Later in [Bre59], J. L. Brenner used minors of the diagonally dominant matrix to give a lower bound on the determinant of that matrix and mentioned that the results can hold for permanents. We have given a proof for that in Theorem 3.5. In [MN62] by Marcus and Newman, the main focus was to observe the behaviour of convergence of the sequence $\{(\text{per}(A^m))^{\frac{1}{m}}\}_m$ as limit $m \rightarrow \infty$ where $A_{n \times n}$ is positive semi-definite hermitian matrix. We have explored how they deduce the permanent from tensor product in Section 4 and observed that a conjecture they have given in [MN62] is a consequence of Theorem 3.6 and gave it as Theorem 4.2. Our approach was mostly combinatorial. Uma Girish showed an 2^n -approximate algorithm to calculate the permanent of laplacian of a simple graph with n vertices in her Master's Thesis [Gir18]. But Bapat's Theorem 5.1 motivated us that we might improve that bound even further. We looked for closed form expressions for permanent of laplacian of paths, circles and trees based on Bapat's result. Recently Hu's and Kalaso's work [HK21] shows that permanent of laplacian of a bipartite pfaffian graph G is equal to the determinant of laplacian of its pfaffian oriented graph which we described in Section 6.

2 Notations and basic definitions

Let us first describe what we mean by laplacian of graphs and permanents of a matrix along with notations that we will use throughout this paper.

We have a simple graph $G = (V, E)$. Assume $V = \{v_1, v_2, \dots, v_n\}$ and $E = \{e_1, e_2, \dots, e_m\}$ be the set of vertices and set of edges of the graph G , respectively. We denote the adjacency matrix by

$A = (a_{ij})_{n \times n}$, diagonal matrix by $B = (b_{ij})_{n \times n}$ and incidence matrix by $Q = (q_{ij})_{n \times n}$, where

$$a_{ij} = \begin{cases} 1, & \text{if } v_i \text{ and } v_j \text{ are adjacent} \\ 0, & \text{Otherwise} \end{cases} \quad (1)$$

$$b_{ij} = \begin{cases} \deg(v_i) = d_i, & \text{if } i = j \\ 0, & \text{Otherwise} \end{cases} \quad (2)$$

$$q_{ij} = \begin{cases} 1, & \text{if } e_j \text{ originates } v_i \\ -1, & \text{if } e_j \text{ terminates at } v_i \\ 0, & \text{Otherwise} \end{cases} \quad (3)$$

Then laplacian of G defined by $L = B - A$, where L is symmetric positive semi-definite. A simple observation gives us that $L = QQ^t$.

Let $K \subset \{1, 2, \dots, n\}$. Let $H(K)$ be the set of all sub-graphs S of G satisfying the following properties

- $V(S) = \{v_i : i \in K\}$
- Any connected component of H will either be an edge or a cycle

In general any sub-graph S of G with above two properties, is called Sachs sub-graph of G . We define $c(S)$, $c_0(S)$, $c^+(S)$ by the number of components of S which are cycles, the number of odd cycles contained in S , the number of even cycles contained in S , respectively.

We have permanent of matrix $A_{n \times n}$ as follows

$$\text{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)} = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{\sigma(i), i} = \frac{1}{n!} \sum_{\sigma, \tau \in S_n} \prod_{i=1}^n a_{\tau(i) \sigma(i)}$$

We have determinant of matrix $A_{n \times n}$ as follows,

$$\det(A) = \sum_{\sigma \in S_n} \text{sign}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}$$

Given a matrix $A_{p \times n} = (a_{ij})_{p \times n}$, we define $A_{[c_1, c_2, \dots, c_p]}^{r_1, r_2, \dots, r_p}$ as the sub-matrix formed by rows $r_1, r_2, \dots, r_p$ of A and columns $c_1, c_2, \dots, c_p$ of A . Also we define $A_{\{i_1, i_2, \dots, i_l\} \times \{j_1, j_2, \dots, j_l\}}$ as the sub-matrix formed by removing the rows $i_1, i_2, \dots, i_l$ of A and columns $j_1, j_2, \dots, j_l$ of A . Set $O_a(b, c) = \{(t_1, t_2, \dots, t_a) : b < t_1 < t_2 < \dots < t_a \leq c \text{ and } t_l \in \mathbb{Z}, \forall l \in \{1, 2, \dots, a\}\}$.

Later in Section 4, we would like to visualise the permanent function as a special case of inner product of two vectors from a tensor product space under some suitable conditions. Suppose U

be an n - dimensional unitary vector space. Let $U^{(k)}$ be the space of k tensors on U in which the k -tensor product of any $x_1, x_2, \dots, x_k \in U$ is defined as follows

$$x_1 \otimes x_2 \otimes \dots \otimes x_k(\phi) = \phi(x_1, x_2, \dots, x_k), \forall \phi \in L(U^k, \mathbb{C})$$

where $L(U^k, \mathbb{C})$ is the set of all multi-linear complex valued functions of vectors of $U^k = \underbrace{U \times U \times \dots \times U}_{k \text{ times}}$.

Let G_p be a subgroup of S_k with $|G_p| = p$ and c be a character of degree 1 in G_p . Now we define $T : U^{(k)} \rightarrow U^{(k)}$ by

$$T(x_1 \otimes x_2 \otimes \dots \otimes x_k) = \frac{1}{p} \sum_{\mu \in G_p} c(\mu) x_{\mu(1)} \otimes x_{\mu(2)} \otimes \dots \otimes x_{\mu(k)}$$

The inner product on $U^{(k)}$ is defined by

$$(x_1 \otimes x_2 \otimes \dots \otimes x_k, y_1 \otimes y_2 \otimes \dots \otimes y_k) = \prod_{i=1}^k \underbrace{(x_i, y_i)}_{\text{inner product in } U}$$

Thus one easy observation is that T is idempotent and hermitian with respect to the above inner product on $U^{(k)}$.

Afterwards we will introduce the pfaffian graphs along with several related results for permanents in Section 6.

3 Similarities between permanent and determinant

Theorem 3.1 (Frobenius-Konig theorem). *The permanent of a non-negative $A_{n \times n}$ matrix is 0 if and only if it has a zero sub-matrix of order $r \times s$ with $r + s = n + 1$*

Lemma 3.2 (Binet and Cauchy formula for Determinants). $\det(AB^t) = \sum_{\alpha \in G_{n,m}} \det(A(\alpha)) \det(B(\alpha))$

where A, B are of $n \times m$ orders. Here $G_{n,m}$ is the set of all strictly increasing sequences $\alpha = (\alpha_1, \dots, \alpha_n)$ of n -tuples, where $\alpha_i \in \{1, 2, \dots, m\}$, and $A(\alpha)$ and $B(\alpha)$ are $n \times n$ sub-matrices with i -th column being chosen $\alpha(i)$ times from A and B , respectively.

Similarly we have a formula for permanents as follows

Lemma 3.3 (Binet and Cauchy formula for Permanents). $\text{per}(AB^t) = \sum_{\alpha \in G_{n,m}} \frac{1}{\alpha!} \text{per}(A(\alpha)) \text{per}(B(\alpha))$

where A, B are of $n \times m$ orders. Here $G_{n,m}$ is the set of all non-decreasing sequences $\alpha = (\alpha_1, \dots, \alpha_n)$ of n -tuples, where $\alpha_i \in \{1, 2, \dots, m\}$.

Ans: Let $AB^t = (x_{ij})_{n \times n}$. Now we have,

$$\begin{aligned}
\text{per}(AB^t) &= \frac{1}{n!} \sum_{\sigma, \tau \in S_n} \prod_{i=1}^n x_{\tau(i)\sigma(i)} \\
&= \frac{1}{n!} \sum_{\sigma, \tau \in S_n} \prod_{i=1}^n \sum_{k=0}^m a_{\tau(i)k} b_{\sigma(i)k} \\
&= \frac{1}{n!} \sum_{\sigma, \tau \in S_n} \sum_{f: [m] \rightarrow [n]} \prod_{i=1}^m a_{\tau(i)f(i)} \prod_{i=1}^n b_{\sigma(i)f(i)} \\
&= \frac{1}{n!} \sum_{f: [n] \rightarrow [m]} \sum_{\sigma, \tau \in S_n} \prod_{i=1}^n a_{\tau(i)f(i)} \prod_{i=1}^n b_{\sigma(i)f(i)} \\
&= \frac{1}{n!} \sum_{f: [n] \rightarrow [m]} \sum_{\tau \in S_n} \prod_{i=1}^n a_{\tau(i)f(i)} \sum_{\sigma \in S_n} \prod_{i=1}^n b_{\sigma(i)f(i)} \\
&= \sum_{\alpha \in \{0,1,2,\dots,m\}^n \text{ such that } \sum_i \alpha(i)=n} \frac{1}{\alpha!} \text{per}A(\alpha) \text{per}B(\alpha)
\end{aligned}$$

Here, $A(\alpha)$, $B(\alpha)$ is $n \times n$ sub-matrix with i -th column being chosen $\alpha(i)$ times from A and B , respectively.

Theorem 3.4 ([Bre59, Theorem 1]). Given a matrix $A_{p \times n} = (a_{ij})_{p \times n}$ with $p < n$ and satisfying the following properties,

- $a_{ii} \neq 0, \forall i \in \{1, 2, \dots, p\}$
- $\sigma_i |a_{ii}| = \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|$, where $\sigma_i \in [0, 1] \forall i \in \{1, 2, \dots, p\}$.

Assume $p = r_0 \leq r_1 \leq \dots \leq r_{p-1} \leq r_p = n$ be fixed integers. Then we have

$$\sigma_1 \sigma_2 \dots \sigma_p \left| \det \left(A_{[1,2,\dots,p]}^{[1,2,\dots,p]} \right) \right| \geq \sum_{s=1}^p \sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} \left| \det \left(A_{[1,2,\dots,(p-s), t_1, t_2, \dots, t_s]}^{[1,2,\dots,p]} \right) \right|$$

J.L. Brenner mentioned in paper [Bre59] that this theorem can be extended to permanent. We have given the permanent version of above theorem with proof.

Theorem 3.5. Given a matrix $A_{p \times n} = (a_{ij})_{p \times n}$ with $p < n$ and satisfying the following properties,

- $a_{ii} \neq 0, \forall i \in \{1, 2, \dots, p\}$
- $\sigma_i |a_{ii}| = \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|$, where $\sigma_i \in [0, 1] \forall i \in \{1, 2, \dots, p\}$.

Assume $p = r_0 \leq r_1 \leq \dots \leq r_{p-1} \leq r_p = n$ be fixed integers. Then we have

$$\sigma_1 \sigma_2 \dots \sigma_p |\text{per}(A[{}^{1,2,\dots,p}_{1,2,\dots,p}])| \geq \sum_{s=1}^p [\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} |\text{per}(A[{}^{1,2,\dots,p}_{1,2,\dots,(p-s),t_1,t_2,\dots,t_s}])|]$$

Here empty sums are 0 and empty products are 1.

Proof: We have a matrix $A_{p \times n} = (a_{ij})_{p \times n}$ with $p < n$ and satisfying the following properties,

1. $a_{ii} \neq 0, \forall i \in \{1, 2, \dots, p\}$
2. $\sigma_i |a_{ii}| = \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|$, where $\sigma_i \in [0, 1] \forall i \in \{1, 2, \dots, p\}$.

For $p = 1$, the result is obvious. We will going to use strong induction to prove the theorem. Let us assume statement of the theorem is true for matrices with $p - 1$ rows. Assume $p = r_0 \leq r_1 \leq \dots \leq r_{p-1} \leq r_p = n$ be fixed integers. Now let us start from the following expression

$$\begin{aligned} & \left| \text{per}(A[{}^{1,2,\dots,p}_{1,2,\dots,(p-s),t_1,t_2,\dots,t_s}]) \right| \\ &= \left| \sum_{\substack{f: \{1,2,\dots,p\} \xrightarrow{\text{bijective}} \{1,2,\dots,(p-s),t_1,t_2,\dots,t_s\}}} \prod_{i=1}^p a_{if(i)} \right| \\ &= \left| \sum_{\substack{f: \{1,2,\dots,p\} \xrightarrow{\text{bijective}} \{1,2,\dots,(p-s),t_1,t_2,\dots,t_s\}}} a_{pf(p)} \prod_{i=1}^{(p-1)} a_{if(i)} \right| \end{aligned}$$

Now we will write the sum with respect to entries of the p -th row involed in the summation

$$\begin{aligned} &= \left| \sum_{q \in \{1,2,\dots,(p-s),t_1,t_2,\dots,t_s\}} a_{pq} \sum_{\substack{g: \{1,2,\dots,(p-1)\} \xrightarrow{\text{bijective}} \{1,2,\dots,(p-s),t_1,t_2,\dots,t_s\} \setminus \{q\}}} \prod_{i=1}^{(p-1)} a_{ig(i)} \right| \\ &= \left| \sum_{q \in \{1,2,\dots,(p-s),t_1,t_2,\dots,t_s\}} a_{pq} \text{per}(A[{}^{1,2,\dots,(p-1)}_{1,2,\dots,(p-s),t_1,t_2,\dots,t_s \setminus \{q\}}]) \right| \\ &= \left| \sum_{q=1}^{p-s} a_{pq} \text{per}(A[{}^{1,2,\dots,(p-1)}_{1,2,\dots,(p-s),t_1,t_2,\dots,t_s \setminus \{q\}}]) \right| + \left| \sum_{q \in \{t_1,t_2,\dots,t_s\}} a_{pq} \text{per}(A[{}^{1,2,\dots,(p-1)}_{1,2,\dots,(p-s),t_1,t_2,\dots,t_s \setminus \{q\}}]) \right| \\ &\leq \sum_{q=1}^{p-s} |a_{pq}| \left| \text{per}(A[{}^{1,2,\dots,(p-1)}_{1,2,\dots,(p-s),t_1,t_2,\dots,t_s \setminus \{q\}}]) \right| + \sum_{q \in \{t_1,t_2,\dots,t_s\}} |a_{pq}| \left| \text{per}(A[{}^{1,2,\dots,(p-1)}_{1,2,\dots,(p-s),t_1,t_2,\dots,t_s \setminus \{q\}}]) \right| \\ &\leq A + B \end{aligned}$$

where

$$A = \sum_{q=1}^{p-s} |a_{pq}| \left| \text{per}(A_{[1,2,\dots,(p-s),t_1,t_2,\dots,t_s]\setminus\{q\}}^{1,2,\dots,(p-1)}) \right|,$$

$$B = \sum_{q \in \{t_1,t_2,\dots,t_s\}} |a_{pq}| \left| \text{per}(A_{[1,2,\dots,(p-s),t_1,t_2,\dots,t_s]\setminus\{q\}}^{1,2,\dots,(p-1)}) \right|$$

Now we look at the following

$$\begin{aligned} & \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} \left| \text{per}(A_{[1,2,\dots,(p-s),t_1,t_2,\dots,t_s]}^{1,2,\dots,p}) \right| \right) \\ & \leq \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} (A + B) \right) \\ & = \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} A \right) + \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} B \right) \end{aligned}$$

We will see each of the terms of R.H.S. in above separately. The first term in above can be analysed as below

$$\begin{aligned} & \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} A \right) \\ & = \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} \sum_{q=1}^{p-s} |a_{pq}| \left| \text{per}(A_{[1,2,\dots,(p-s),t_1,t_2,\dots,t_s]\setminus\{q\}}^{1,2,\dots,(p-1)}) \right| \right) \\ & = \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{q=1}^{p-s} |a_{pq}| \sum_{t \in O_s(r_{s-1}, r_s)} \left| \text{per}(A_{[1,2,\dots,(p-s),t_1,t_2,\dots,t_s]\setminus\{q\}}^{1,2,\dots,(p-1)}) \right| \right) \\ & = \sum_{q=1}^{p-1} |a_{pq}| \sum_{s=1}^{p-q} \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} \left| \text{per}(A_{[1,2,\dots,(p-s),t_1,t_2,\dots,t_s]\setminus\{q\}}^{1,2,\dots,(p-1)}) \right| \right) = C \end{aligned}$$

The last term here can be analysed as below

$$\begin{aligned} & \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} B \right) \\ &= \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} \sum_{q \in \{t_1, t_2, \dots, t_s\}} |a_{pq}| \left| \text{per}(A_{[1, 2, \dots, (p-s), t_1, t_2, \dots, t_s] \setminus \{q\}}^{1, 2, \dots, (p-1)}) \right| \right) \end{aligned}$$

Now we can add more non-negative terms as we move from s to $s - 1$ length sequences.

$$\begin{aligned} & \leq \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \left[\sum_{q=1+r_{s-1}}^{r_s} |a_{pq}| \right] \sum_{t \in O_{s-1}(r_{s-1}, r_s)} \left| \text{per}(A_{[1, 2, \dots, (p-s), t_1, t_2, \dots, t_{s-1}] }^{1, 2, \dots, (p-1)}) \right| \right) \\ & \leq \sum_{s=1}^p \left(\left[\sum_{q=1+r_{s-1}}^{r_s} |a_{pq}| \right] \sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_{s-1}(r_{s-1}, r_s)} \left| \text{per}(A_{[1, 2, \dots, (p-s), t_1, t_2, \dots, t_{s-1}] }^{1, 2, \dots, (p-1)}) \right| \right) \end{aligned}$$

Using strong induction hypothesis for matrix $A_{(p-1) \times n}$

$$\begin{aligned} & \leq \sum_{s=1}^p \left(\left[\sum_{q=1+r_{s-1}}^{r_s} |a_{pq}| \right] \sigma_1 \sigma_2 \dots \sigma_{p-1} \left| \text{per}(A_{[1, 2, \dots, (p-1)] }^{1, 2, \dots, (p-1)}) \right| \right) \\ & \leq \sigma_1 \sigma_2 \dots \sigma_{p-1} \left| \text{per}(A_{[1, 2, \dots, (p-1)] }^{1, 2, \dots, (p-1)}) \right| \sum_{s=1}^p \left(\left[\sum_{q=1+r_{s-1}}^{r_s} |a_{pq}| \right] \right) \\ & \leq \sigma_1 \sigma_2 \dots \sigma_{p-1} \left| \text{per}(A_{[1, 2, \dots, (p-1)] }^{1, 2, \dots, (p-1)}) \right| \left(\sum_{q=1+p}^n |a_{pq}| \right) \end{aligned}$$

Using the property 2 of given matrix in the theorem, we get

$$\begin{aligned} & = \sigma_1 \sigma_2 \dots \sigma_{p-1} \left| \text{per}(A_{[1, 2, \dots, (p-1)] }^{1, 2, \dots, (p-1)}) \right| \left(\sigma_p |a_{pp}| - \sum_{q=1}^{p-1} |a_{pq}| \right) \\ & \leq \sigma_1 \sigma_2 \dots \sigma_p \left| \text{per}(A_{[1, 2, \dots, p]}^{1, 2, \dots, p}) \right| - \left(\sigma_1 \sigma_2 \dots \sigma_{p-1} \sum_{q=1}^{p-1} |a_{pq}| \right) \left(\left| \text{per}(A_{[1, 2, \dots, (p-1)] }^{1, 2, \dots, (p-1)}) \right| - \left| \text{per}(A_{[1, 2, \dots, p] \setminus \{q\}}^{1, 2, \dots, (p-1)}) \right| \right) \\ & = \sigma_1 \sigma_2 \dots \sigma_p \left| \text{per}(A_{[1, 2, \dots, p]}^{1, 2, \dots, p}) \right| - D \end{aligned}$$

where

$$D = \sigma_1 \sigma_2 \dots \sigma_{p-1} \sum_{q=1}^{p-1} |a_{pq}| \left(\left| \text{per}(A_{[1, 2, \dots, (p-1)] }^{1, 2, \dots, (p-1)}) \right| - \left| \text{per}(A_{[1, 2, \dots, p] \setminus \{q\}}^{1, 2, \dots, (p-1)}) \right| \right)$$

Now combining the inequalities from above, we get

$$\begin{aligned}
& \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} \left| \text{per}(A_{[1,2,\dots,(p-s), t_1, t_2, \dots, t_s]}) \right| \right) \\
& \leq \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} A \right) + \sum_{s=1}^p \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} B \right) \\
& \leq C + \sigma_1 \sigma_2 \dots \sigma_p \left| \text{per}(A_{[1,2,\dots,p]}) \right| - D
\end{aligned}$$

Therefore, if we can show that $D \geq C$ then our proof will be complete. Now to do that we will again use induction hypothesis for matrix $A_{(p-1) \times n}$. Let us first define $\{v_i\}_{i=0}^{p-1}$ as $v_0 = p-1$, $v_i = r_i$ for $i = 1, 2, \dots, p-2$, and $v_{p-1} = n$. Then clearly $v_0 \leq v_1 \leq v_2 \leq \dots \leq v_{p-1}$. Thus using this fixed integers $\{v_i\}_{i=0}^{p-1}$ and for each $q \in \{1, 2, \dots, (p-1)\}$, we can use induction hypothesis as follows

$$\begin{aligned}
& \sigma_1 \sigma_2 \dots \sigma_{p-1} \left| \text{per}(A_{[1,2,\dots,(p-1)]}) \right| \geq \sum_{s=1}^{p-q} \sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(v_{s-1}, v_s)} \left| \text{per}(A_{[1,2,\dots,(p-s)] \setminus \{q\}, t_1, t_2, \dots, t_s}) \right| \\
& \geq \sum_{s=1}^{p-q} \sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} \left| \text{per}(A_{[1,2,\dots,(p-s)] \setminus \{q\}, t_1, t_2, \dots, t_s}) \right| + \sigma_1 \sigma_2 \dots \sigma_{p-1} \left| \text{per}(A_{[1,2,\dots,p] \setminus \{q\}}) \right|
\end{aligned}$$

Here, we considered first term separately from the summation in R.H.S.

$$\begin{aligned}
& \Rightarrow \sigma_1 \sigma_2 \dots \sigma_{p-1} [|\text{per}(A_{[1,2,\dots,(p-1)]})| - |\text{per}(A_{[1,2,\dots,p] \setminus \{q\}})|] \\
& \geq \sum_{s=1}^{p-q} \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} |\text{per}(A_{[1,2,\dots,(p-s)] \setminus \{q\}, t_1, t_2, \dots, t_s})| \right)
\end{aligned}$$

Now taking sum over $|a_{pq}|, \forall q \in \{1, 2, \dots, p-1\}$ in both sides we get

$$\begin{aligned}
D & \geq \sum_{q=1}^{p-1} |a_{pq}| \sum_{s=1}^{p-q} \left(\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} |\text{per}(A_{[1,2,\dots,(p-s)] \setminus \{q\}, t_1, t_2, \dots, t_s})| \right) = C \\
& \Rightarrow D \geq C
\end{aligned}$$

Thus proved.

Now if we restrict the domain of σ_i i.e. $0 \leq \sigma_i < 1, \forall i \in \{1, 2, \dots, p\}$, then the inequality can be strict in Theorem 3.5 and we will have the following theorem

Theorem 3.6. Given a matrix $A_{p \times n} = (a_{ij})_{p \times n}$ with $p < n$ and satisfying the following properties,

- $a_{ii} \neq 0, \forall i \in \{1, 2, \dots, p\}$

$$\bullet \sigma_i |a_{ii}| = \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|, \text{ where } \sigma_i \in [0, 1) \forall i \in \{1, 2, \dots, p\}.$$

Assume $p = r_0 \leq r_1 \leq \dots \leq r_{p-1} \leq r_p = n$ be fixed integers. Then we have

$$\sigma_1 \sigma_2 \dots \sigma_p |\text{per}(A_{[1,2,\dots,p]}^{1,2,\dots,p})| > \sum_{s=1}^p [\sigma_1 \sigma_2 \dots \sigma_{p-s} \sum_{t \in O_s(r_{s-1}, r_s)} |\text{per}(A_{[1,2,\dots,(p-s), t_1, t_2, \dots, t_s]}^{1,2,\dots,p})|]$$

Here empty sums are 0 and empty products are 1.

Proof: The proof will proceed exactly same as Theorem 3.5 except in those steps where we have used induction hypotheses. Due to the induction hypotheses we will get strict inequality and we will have $D - C > 0$ as $\sigma(i) < 1$.

4 The permanent function from tensor product

We are given an n -dimensional unitary vector space U . G_p be a subgroup of S_k with $|G_p| = p$ and c be a character of degree 1 in G_p . Recall from Section 2, $U^{(k)}$ is the space of k tensors on U in which the k -tensor product of any $x_1, x_2, \dots, x_k \in U$ is defined by

$$x_1 \otimes x_2 \otimes \dots \otimes x_k(\phi) = \phi(x_1, x_2, \dots, x_k), \forall \phi \in L(U^k, \mathbb{C})$$

The idempotent and hermitian operator $T : U^{(k)} \rightarrow U^{(k)}$ is given by

$$T(x_1 \otimes x_2 \otimes \dots \otimes x_k) = \frac{1}{p} \sum_{\sigma \in G_p} c(\sigma) x_{\sigma(1)} \otimes x_{\sigma(2)} \otimes \dots \otimes x_{\sigma(k)}$$

Theorem 4.1 ([MN62, Theorem 1]). *If $\{x_i\}_{i=1}^k, \{y_i\}_{i=1}^k$ be vectors in U , then we have*

$$(T(x_1 \otimes x_2 \otimes \dots \otimes x_k), T(y_1 \otimes y_2 \otimes \dots \otimes y_k)) = \frac{1}{p} \sum_{\sigma \in G_p} c(\sigma) \left(\prod_{i=1}^k (x_{\sigma(i)}, y_i) \right)$$

Proof: From definition of T , we have

$$\begin{aligned}
& (T(x_1 \otimes x_2 \otimes \cdots \otimes x_k), T(y_1 \otimes y_2 \otimes \cdots \otimes y_k)) \\
&= (T^*T(x_1 \otimes x_2 \otimes \cdots \otimes x_k), y_1 \otimes y_2 \otimes \cdots \otimes y_k) \\
&\text{now we will use the fact that } T^*T = T^2 = T \\
&= (T(x_1 \otimes x_2 \otimes \cdots \otimes x_k), y_1 \otimes y_2 \otimes \cdots \otimes y_k) \\
&= \left(\frac{1}{p} \sum_{\sigma \in G_p} c(\sigma) x_{\sigma(1)} \otimes x_{\sigma(2)} \otimes \cdots \otimes x_{\sigma(k)}, y_1 \otimes y_2 \otimes \cdots \otimes y_k\right) \\
&= \frac{1}{p} \sum_{\sigma \in G_p} c(\sigma) (x_{\sigma(1)} \otimes x_{\sigma(2)} \otimes \cdots \otimes x_{\sigma(k)}, y_1 \otimes y_2 \otimes \cdots \otimes y_k) \\
&= \frac{1}{p} \sum_{\sigma \in G_p} c(\sigma) \left(\prod_{i=1}^k (x_{\sigma(i)}, y_i) \right)
\end{aligned}$$

4.1 Observation

- Take $G_p = S_k$, and $c(\sigma) = \text{sign}(\sigma)$, $\forall \sigma \in G_p$, and $T(x_1 \otimes x_2 \otimes \cdots \otimes x_k) = x_1 \wedge x_2 \wedge \cdots \wedge x_k$ is the grassmann product. Then Theorem 4.1 gives

$$(x_1 \wedge x_2 \wedge \cdots \wedge x_k, y_1 \wedge y_2 \wedge \cdots \wedge y_k) = \frac{1}{k!} \det((x_i, y_i))$$

- Take $G_p = S_k$, and $c(\sigma) = 1$, $\forall \sigma \in G_p$, and $T(x_1 \otimes x_2 \otimes \cdots \otimes x_k) = x_1 \wedge x_2 \wedge \cdots \wedge x_k$ is the grassmann product. Then Theorem 4.1 gives

$$(x_1 \wedge x_2 \wedge \cdots \wedge x_k, y_1 \wedge y_2 \wedge \cdots \wedge y_k) = \frac{1}{k!} \text{per}((x_i, y_i))$$

One of the conjectures in paper [MN62] was the following

1. $\text{per}(I - A) \geq 0$, $\forall A \in \Omega_n$, where Ω_n = set of all non-negative doubly stochastic matrices.

Marcus showed the above conjectures only for symmetric matrices in Ω_n [MN62]. But from brenner's Theorem 3.6 it followed easily.

Theorem 4.2. $\text{per}(I - A) \geq 0$, $\forall A \in \Omega_n$, where Ω_n = set of all non-negative doubly stochastic matrices.

Proof: Now take any $A \in \Omega_n$. Then we have $\sum_{j=1}^n a_{ij} = 1$, $\forall i$. Define $B_\lambda = I - \lambda A$, $\forall \lambda \in [0, 1)$.

Let us look at entries of $B_\lambda = (b_{ij})_{n \times n}$, where $b_{ij} = \begin{cases} 1 - \lambda a_{ii}, & \text{for } i = j \\ -\lambda a_{ij}, & \text{for } i \neq j \end{cases}$.

Now look at i^{th} row of B_λ ,

$$\sum_{j=1, j \neq i}^n |b_{ij}| = \sum_{j=1, j \neq i}^n \lambda |a_{ij}| = \lambda \sum_{j=1, j \neq i}^n |a_{ij}| = \lambda(1 - a_{ii})$$

Also

$$|b_{ii}| = |1 - \lambda a_{ii}| = 1 - \lambda a_{ii} > 0, (\text{since } a_{ii} \leq 1, \lambda < 1 \implies \lambda a_{ii} < 1)$$

Thus if we look for σ_i for which we can have the following equation

$$\sigma_i |b_{ii}| = \sum_{j=1, j \neq i}^n |b_{ij}|$$

$$\text{i.e. if, } \sigma_i(1 - \lambda a_{ii}) = \lambda(1 - a_{ii})$$

Note $\sigma_i = 1$ iff $\lambda = 1$ but $\lambda \in [0, 1) \implies \sigma \in [0, 1)$.

Thus we can use Theorem 3.6, which gives us for each $\lambda \in [0, 1)$, we have $|\text{per}(B_\lambda)| > 0 \implies |\text{per}(I - \lambda A)| > 0$. Since $\text{per}(B_\lambda)$ is continuous function with respect to λ , we must have either $\text{per}(I - \lambda A) > 0$ or $\text{per}(I - \lambda A) < 0$ for all $\lambda \in [0, 1)$. Note for $\lambda = 0$ $\text{per}(B_\lambda) = \text{per}(I) = 1$ which implies $\text{per}(I - \lambda A) > 0$ for all $\lambda \in [0, 1)$.

we claim that for $\lambda = 1$, we must have $\text{per}(I - A) \geq 0$ otherwise if $\text{per}(I - A) < 0$ then using continuity, we must have one $\lambda \in (0, 1)$ such that $\text{per}(I - \lambda A) = 0$ [Contradiction]. Thus we proved for any $A \in \Omega_n$, $\text{per}(I - A) \geq 0$.

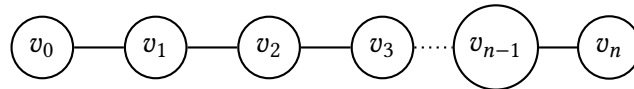
The easy observation is that A need not be doubly stochastic to hold this proof.

5 Approach to compute the permanents of graph laplacians efficiently

We will going to discuss about permanent of laplacian of planar graph G . We will try to achieve efficient algorithm to estimate the permanents. Let us analyse few easy cases where we can achieve a polynomial time algorithm to estimate permanent of graph laplacian.

5.1 Graph is a Path

Suppose we have a path P on $n + 1$ vertices. Say $V = \{v_0, v_1, \dots, v_n\}$



Let us denote $\text{per}(L(P)_{(n+1) \times (n+1)})$ by $T(n+1)$. For any path P of length n the Laplacian of that path will look like

$$T(n+1) = L(P)_{(n+1) \times (n+1)} = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & -1 & \cdots \\ \vdots & \vdots & \ddots & \vdots & -1 \\ 0 & \cdots & 0 & -1 & 1 \end{bmatrix}$$

Let us define a matrix $S_n = \{s_{ij}\}_{n \times n}$ as follows

$$s_{ij} = \begin{cases} 2, & \text{if } i = j \text{ and } i, j < n \\ 1, & \text{if } i = j = n \\ -1, & \text{if } i = j + 1 \text{ or } i = j - 1 \\ 0, & \text{Otherwise} \end{cases}$$

The matrix form looks like

$$S_n = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & -1 & \cdots \\ \vdots & \vdots & \ddots & \vdots & -1 \\ 0 & \cdots & 0 & -1 & 1 \end{bmatrix}$$

Now consider $\text{per}(L(P)_{(n+1) \times (n+1)})$ and expanding along first row we get

$$\text{per}(L(P)_{(n+1) \times (n+1)}) = \text{per}(L(P)_{\{1\} \times \{1\}}) - \text{per}(L(P)_{\{1\} \times \{2\}})$$

now expanding second term with respect to 1st coloumn, we get from above

$$= \text{per}(L(P)_{\{1\} \times \{1\}}) - (-\text{per}(L(P)_{\{1,2\} \times \{1,2\}}))$$

$$= \text{per}(S_n) + \text{per}(S_{n-1}).$$

Now note that $\text{per}(S_1) = 1, \text{per}(S_2) = 3$. Similarly if we expand $\text{per}(S_{n+1})$, we get for each $n \geq 1$

$$\text{per}(S_{n+1}) = 2 \text{per}(S_n) + \text{per}(S_{n-1})$$

Also we have for each $n \geq 1$,

$$T(n+1) = \text{per}(L(P)_{(n+1) \times (n+1)}) = \text{per}(S_n) + \text{per}(S_{n-1})$$

Combining above two recursions, we get

$$T(n) = 2 T(n-1) + T(n-2), \forall n \geq 2$$

with $T(0) = 0, T(1) = 2$.

This is a linear homogeneous recurrence relation of the second order and its solution has the form $T(n) = cr^n$ for $c, r \neq 0, n \geq 1$. After substitution we get $cr^n = 2cr^{n-1} + cr^{n-2}$. The characteristic equation that we obtain is $r^2 - 2r - 1 = 0$. The roots of this equation are $r = (1 \pm \sqrt{2})$. So, the general solution is $F_n = c_1[(1+\sqrt{2})]^n + c_2[(1-\sqrt{2})]^n$. By using initial conditions $T(0) = 0, T(1) = 2$ we can determine the constants c_1 and c_2 . Their values are $c_1 = 1/\sqrt{2}$ and $c_2 = -1/\sqrt{2}$. Therefore, the general form of this recurrence relation is $T(n) = 1/\sqrt{2}[(1+\sqrt{2})]^n - 1/\sqrt{2}[(1-\sqrt{2})]^n$.

5.2 Graph is a Cycle

Let us move on to another simple structure known as cycles. Look at the following: Let C_n be the cycle on n vertices. Denote $L(C_n)$ be the laplacian matrix of C_n .

$$L(C)_{n \times n} = \begin{bmatrix} 2 & -1 & 0 & \cdots & -1 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & -1 & \cdots \\ \vdots & \vdots & \ddots & \vdots & -1 \\ -1 & \cdots & 0 & -1 & 2 \end{bmatrix}$$

Let us define two different types of matrices $F_n = \{f_{ij}\}_{n \times n}$ and $H_n = \{h_{ij}\}_{n \times n}$ as follows

$$f_{ij} = \begin{cases} 2, & \text{if } i = j \\ -1, & \text{if } i = j + 1 \text{ or } i = j - 1 \\ 0, & \text{Otherwise} \end{cases}$$

and

$$h_{ij} = \begin{cases} -1, & \text{if } i = j \\ 2, & \text{if } j = i + 1 \\ -1, & \text{if } j = i + 2 \\ 0, & \text{Otherwise} \end{cases}$$

In matrix form, F_n and H_n looks like

$$F_n = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & -1 & \cdots \\ \vdots & \vdots & \ddots & \vdots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}, H_n = \begin{bmatrix} -1 & 2 & -1 & 0 \cdots & 0 \\ 0 & -1 & 2 & -1 \cdots & 0 \\ 0 & 0 & -1 & 2 & \cdots \\ \vdots & \vdots & \ddots & \vdots & 2 \\ 0 & \cdots & 0 & 0 & -1 \end{bmatrix}$$

Observe that F_n is the laplacian of a path on n vertices having one self loop at each of the end vertices of that path. Now let us expand $\text{per}(L(C)_{n \times n})$ with respect to the first row, then we have

$$\text{per}(L(C)_{n \times n}) = 2 \text{per}(L(C)_{\{1\} \times \{1\}}) - \text{per}(L(C)_{\{1\} \times \{2\}}) - \text{per}(L(C)_{\{1\} \times \{n\}})$$

now expanding second term and third term with respect to 1st column and last row of respective matrices, we get ,

$$\begin{aligned} &= 2 \text{per}(F_{n-1}) - (-\text{per}(L(C)_{\{1,2\} \times \{1,2\}}) - \text{per}(L(C)_{\{1,n\} \times \{1,2\}})) - (-\text{per}(L(C)_{\{1,n\} \times \{1,n\}})) \\ &\quad - (-\text{per}(L(C)_{\{1,n\} \times \{n-1,n\}})) \\ &= 2 \text{per}(F_{n-1}) + 2 \text{per}(F_{n-2}) + 2.(-1)^n \end{aligned}$$

The last term we got because of following observations

$$F_{n-2} = L(C)_{\{1,2\} \times \{1,2\}} = L(C)_{\{1,n\} \times \{1,n\}}$$

and

$$L(C)_{\{1,n\} \times \{1,2\}} = H_{n-2}^T, L(C)_{\{1,n\} \times \{n-1,n\}} = H_{n-2}$$

and due to triangular matrix, we have

$$\text{per}(H_{n-2}) = \text{per}(H_{n-2}^T) = (-1)^{n-2} = (-1)^n$$

Now we need to calculate F_n which is rather simple. $\text{per}(F_1) = 2$, $\text{per}(F_2) = 5$. Also $\text{per}(F_n) = 2 \text{per}(F_{n-1}) + \text{per}(F_{n-2})$, $\forall n \geq 3$. Thus for cycle of length n , we have the linear recursion for permanent of its laplacian as follows,

$$\text{per}(L(C)_{n \times n}) = 2 \text{per}(F_{n-1}) + 2 \text{per}(F_{n-2}) + 2 \cdot (-1)^n$$

with respect to

$$\text{per}(F_n) = 2 \text{per}(F_{n-1}) + \text{per}(F_{n-2}), \forall n \geq 3$$

with boundary condition $\text{per}(F_1) = 2$, $\text{per}(F_2) = 5$. This gives us a polynomial time algorithm to calculate permanent of laplacian for cycles of length n .

Now the next step is to look at trees. But it is not easy to get a linear recursion for the permanent of laplacian of tree just by expanding the permanent of tree-laplacian. Thus we will look at the following result by bapat [Bap86], which will help us to understand the permanent of tree-laplacian differently.

Theorem 5.1 ([Bap86, Theorem 3]). *Let G be a simple graph with n vertices, then*

$$\text{per}(L(G)) = \sum 2^{2c(H) - c^1(H)}$$

where $c(H)$ and $c^1(H)$ denote the number of cycles in H and the number of cycles of length 2 in H , respectively. The summation is over all sub-graphs H of G^2 with n edges satisfying the following properties

- i) In every component of H the number of vertices equals to the number of edges
- ii) There are no cycles of odd length in H .

The first obvious observation is that for simple graph G , $\text{per}(L(G)) \geq 0$, since summation contains all non-negative terms. If G is connected, then we always have one such sub-graph $H \subset G^2$. Let H be spanning tree of G with extra one edge. For this particular sub-graph $c(H) = c^1(H) = 1$, then the corresponding term in the summation is positive and we get $\text{per}(L(G)) > 0$ when G is connected. Also Theorem 5.1 suggests that we might get better bound than 2^n -approximation algorithm described in [Gir18]. To understand the formula 5.1 in detail, the easy way is to look for permanent of laplacian of trees about which we describe in 5.3.

5.3 Permanent of laplacian of a tree

To achieve a close form expression for permanent of laplacian of a tree, let us analyse the Bapat's Theorem 5.1. Suppose we have a tree graph, say, T . We are seeking to calculate the $\text{per}(L(T))$ by the formula in Theorem 5.1. For that we need to calculate the total number of sub-graph $H \subset T^2$ such that in every component of H the number of vertices equals to the number of edges and there are no cycles of odd length in H .

First, note that T^2 is a graph created by doubling the edges of the tree T . Thus T^2 can only have 2 cycles so as any H . Now we have to consider only those H in which every component has the number of vertices equals to the number of edges. If we remove r edges from T , we will get $r + 1$ components, say $T_0, T_1, T_2, T_3, \dots, T_r$ having numbers of vertices $l_0, l_1, l_2, l_3, \dots, l_r$, respectively. Then for each $i \in \{0, 1, 2, \dots, r\}$, T_i will have exactly $l_i - 1$ edges due to the fact that T_i is connected and any more edge will give a cycle in T_i (Note if $l_0 = 1$, then adding one more edge means it will be a self loop) which does not happen for any sub-tree of T . So we can add exactly one more edge to T_i so that it can be a sub-graph of T^2 with required properties. Thus number of ways we can add edge to T_i is $l_i - 1$ ways.

Consider $\Gamma(T^2)$ be set of all sub-graphs of T^2 satisfying two properties given in Theorem 5.1. Let $M(T)$ be the set of all sub-graphs of T . Now define the map

$$\Phi(H) : \Gamma(T^2) \rightarrow M(T)$$

as follows, for any $H \in \Gamma(T^2)$, whenever there are double edges between two vertices in H , it will drop one edge from it. Consider a sub-graph H_k of T by removing k edges from it. i.e H_k will have $k + 1$ components. Now

$$|\Phi^{-1}(H_k)| = \prod_{\substack{C \text{ be connected} \\ \text{components of } H_k}} |E(C)|$$

Here $E(C)$ denotes the set of edges in C . Now the total no of such H_k be

$$\sum_{\substack{S \subset E(T) \\ |S|=k \\ H_k = T \setminus S}} |\Phi^{-1}(H_k)| = \sum_{\substack{S \subset E(T) \\ |S|=k \\ H_k = T \setminus S}} \prod_{\substack{C \text{ be connected} \\ \text{components of } H_k}} |E(C)|$$

. Here each member of $\Phi^{-1}(H_k)$ will have exactly $k + 1$ cycles and all of them are 2-cycle. Thus if we take $k \in \{0, 1, \dots, n - 1\}$, we get exactly the permanent of $L(T)$ by Theorem 5.1

$$\text{per}(L(T)) = \sum_{k=0}^{n-1} 2^{k+1} \sum_{\substack{S \subset E(T) \\ |S|=k \\ H_k = T \setminus S}} |\Phi^{-1}(H_k)| = \sum_{k=0}^{n-1} 2^{k+1} \sum_{\substack{S \subset E(T) \\ |S|=k \\ H_k = T \setminus S}} \prod_{\substack{C \text{ be connected} \\ \text{components of } H_k}} |E(C)| \quad (4)$$

Now we have a compact expression to calculate permanent of laplacian of a tree but it will not run in polynomial time.

Next we are going to look into concepts of pfaffian graphs which will help us to further in approximation of permanents of laplacian of trees.

6 Pfaffian Graphs

Suppose we have an undirected graph G . Let G^λ be an oriented graph obtained from G by orienting each $e \in E(G)$ by a direction. Now take any even cycle C in G^λ . We will say C is oddly oriented if we choose to travel that cycle either clockwise (or anti-clockwise), we will get an odd number of edges having direction towards our travel.

A **Pfaffian Orientation** is an orientation on G such that every even cycle C for which the subgraph $G \setminus C$ has a perfect matching, is oddly oriented after orientation. An undirected graph G is called **Pfaffian** if G has a pfaffian orientation. Thomas [HK21] showed that all planar graphs are pfaffian which was first proved by P.W. Kasteleyn. Since tree is a planar graph, it is pfaffian. Xu and Kalaso [HK21] give result which not only gives a polynomial time algorithm to calculate permanent for trees but also extends to all bipartite planar graphs. The result is given as follows

Theorem 6.1 ([HK21, Theorem 3]). *Let G be a bipartite pfaffian graph of order n and λ be one of its pfaffian orientation. G^λ is the pfaffian oriented graph corresponding to the orientation λ of G . Then*

$$\text{per}(L(G)) = \det(L(G^\lambda))$$

Proof: The proof is shown by observing expressions for $\text{per}(L(G))$ and $\det(L(G^\lambda))$ separately. Recall $c(S)$, $c_0(S)$, $c^+(S)$ for any $S \subset G$ from section 2. Firstly, Brualdi and Goldwasser show that [BG84, lemma 2.1]

$$\text{per}(L(G)) = \sum_{K \subset \{1,2,\dots,n\}} \sum_{S \in H(K)} (-1)^{c_0(S)} 2^{c(S)} \prod_{i \notin K} d_i \quad (5)$$

then Xu and Kalaso [HK21, Theorem 1] show that

$$\det(L(G^\lambda)) = \sum_{K \subset \{1,2,\dots,n\}} \sum_{S^\lambda \in H(K^\lambda)} (-1)^{c^+(S^\lambda)} 2^{c(S^\lambda)} \prod_{i \notin K} d_i \quad (6)$$

Now we are given that G is bipartite and pfaffian. Thus S is pfaffian with S^λ being its corresponding oriented graph. Also S is of even order which means $c_0(S) = 0$. Hence $c^+(S^\lambda) = 0$. Thus R.H.S. of equations 5 and 6 are same (term wise). Hence we have

$$\text{per}(L(G)) = \det(L(G^\lambda))$$

Corollary 6.2. *Theorem 6.1 implies that calculating permanent of laplacian of any tree is equivalent to calculate determinant of laplacian of its pfaffian oriented graph.*

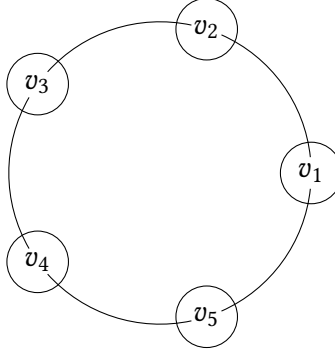
Appendix 1: To understand the 4 in Section 5.3 we tried to simplify it for paths. The observation gives the expression for permanent of laplacian of a path of length n as follows,

$$\text{per}(L(P)) = \sum_{k=0}^{n-1} 2^{k+1} \sum_{\substack{S \subset E(P) \\ |S|=k \\ H_k = P \setminus S}} \prod_{\substack{C \text{ be connected} \\ \text{components of } H_k}} |E(C)| = \sum_{k=0}^{n-1} 2^{k+1} \sum_{\substack{l_1, l_2, \dots, l_{k+1} \in \mathbb{N} \cup \{0\} \\ l_1 + l_2 + \dots + l_{k+1} = n-k}} l_1 l_2 \dots l_{k+1}$$

Though I was not able to prove this.

Appendix 2: We consider a cycle of length n . First approach to calculate the permanent of laplacian of a cycle was that we will use $T(n)$ repeatedly as deleting edges from a cycle will break into several paths. But later, I realised it just gives a rough upper bound on permanent of laplacian of cycles. For now let us analyse the cycle according to odd and even lengths.

Cycle of odd length: Let C be cycle of length $2n + 1$ having vertices $v_1, v_2, \dots, v_{2n+1}$.



for $n = 2$

Claim: For cycle C of length $2n + 1$, We will have

$$\text{per}(L(C)) \leq (2n + 1)T(2n)$$

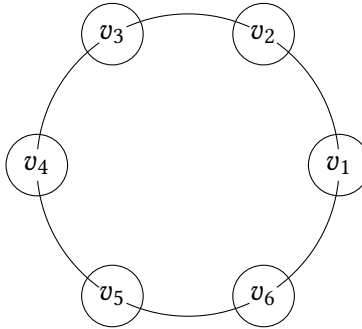
First note that we have to consider all such sub-graphs H of C^2 such that it has no odd cycles and number of edges has to be equal to number of vertices in each connected component of that sub-graph H . We will proceed with the same idea that we are following from starting of Section 4. Let k be the number of edges that we will remove to get components of C .

Now for $k = 0$ implies it is C itself, having number of edges same as number of vertices. But it is an odd cycle of length $2n + 1$. Thus its can not be taken as $H \subset C^2$.

Now for $k = 1$, we are allowed to remove one edges from C which can be done in $2n + 1$ ways. In each case we will get exactly one component which will be a path P of length $2n$ and for all $k \in \{2, 3, \dots, 2n + 1\}$ those cases are already involved in that path P . Thus

$$\text{per}(L(C)) \leq (2n + 1)T(2n)$$

Cycle of even length: Let C be cycle of length $2n$ having vertices $v_1, v_2, \dots, v_{2n}$.



for $n = 3$

Claim: For cycle C of length $2n$, we will have

$$\text{per}(L(C)) \leq 4 + (2n)T(2n - 1)$$

First, note that we have to consider all such sub-graphs H of C^2 such that it has no odd cycles and number of edges has to be equal to number of vertices in each connected component of that sub-graph H . We will proceed with the same idea that we have followed in the previous case. Let k be the number of edges that we will remove to get components of C .

Now for $k = 0$ implies it is C itself, having number of edges same as number of vertices and it is an even cycle of length $2n$. Thus it has to be taken as $H \subset C^2$, which will contribute $2^{(2,1-0)} = 4$ in calculating $\text{per}(L(C))$. Now for $k = 1$, we are allowed to remove one edge from C which can be done in $2n$ ways. In each case we will get exactly one component which will be a path P of length $2n - 1$ and for all $k \in \{2, 3, \dots, 2n\}$, those cases are already involved in that path P . Thus

$$\text{per}(L(C)) \leq 4 + (2n)T(2n - 1)$$

These are the rough upper bounds because of the way we argued, the expression involves contribution of few special sub-graphs more than once.

7 Future work

Through this project work I have couple of problems listed below which I am interested in

1. Meiburg recently showed that within a particular sub-exponential factor it is NP-hard to approximate permanent of hermitian positive semi definite matrix [Mei21]. But we can ask whether one can get any better approximation algorithm than factor 2^n to calculate permanent of laplacian matrix of a given graph $G = (V, E)$ with $|V| = n$.
2. We saw recursive expression for paths and circles. Recursive expression is not known for trees.
3. What can we say about permanent of laplacian of non-planar graphs which are not pfaffian?
4. We have seen that under the deletion of edge of a graph, how permanent of laplacian behaves from graph to its sub-graph [Gir18]. But there is also an interesting property known as "identification" of edges (respectively vertices) within a graph or in between two graphs. What can we say about the behaviour of permanent of laplacian under this operation?

Bibliography

The paper [VY89] gave me the idea of complexity analysis on pfaffian graphs. Zecchina's work [Zec01] motivated me to explore and find whether we can get efficient algorithm to approximate permanent of laplacian of graphs embedded in non-planar surfaces.

References

- [Aar11] Scott Aaronson. A linear-optical proof that the permanent is $\#P$ -hard. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 467(2136):3393–3405, 2011.
- [AGGS17] Nima Anari, Leonid Gurvits, Shayan Oveis Gharan, and Amin Saberi. Simply exponential approximation of the permanent of positive semidefinite matrices. In *2017 IEEE 58th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 914–925. IEEE, 2017.
- [Bap86] R. B. Bapat. A bound for the permanent of the Laplacian matrix. *Linear Algebra and its Applications*, 74:219–223, feb 1986.
- [BG84] Richard A Brualdi and John L Goldwasser. Permanent of the laplacian matrix of trees and bipartite graphs. *Discrete mathematics*, 48(1):1–21, 1984.
- [Bre54] J. L. Brenner. A Bound for a Determinant with Dominant Main Diagonal. *Proceedings of the American Mathematical Society*, 5(4):631–634, 1954.
- [Bre59] J. L. Brenner. Relations among the minors of a matrix with dominant principal diagonal. *Duke Mathematical Journal*, 26(4):563–567, 1959.
- [Gir18] Uma Girish. Deterministic approximation schemes for permanents of laplacians. Master’s thesis, Chennai Mathematical Institute, 2018. Available from <https://www.cs.princeton.edu/~ugirish/>.
- [GKKS16] Ankit Gupta, Pritish Kamath, Neeraj Kayal, and Ramprasad Satharishi. Arithmetic circuits: A chasm at depth 3. *SIAM Journal on Computing*, 45(3):1064–1079, 2016.
- [GS16] Daniel Grier and Luke Schaeffer. New hardness results for the permanent using linear optics. *arXiv preprint arXiv:1610.04670*, 2016.
- [HK21] Xiaoxue Hu and Grace Kalaso. Computing the permanent of the laplacian matrices of nonbipartite graphs. *Journal of Mathematics*, 2021, 2021.
- [JSV04] Mark Jerrum, Alistair Sinclair, and Eric Vigoda. A polynomial-time approximation algorithm for the permanent of a matrix with nonnegative entries. *Journal of the ACM (JACM)*, 51(4):671–697, 2004.
- [LM22] Per Håkan Lundow and Klas Markström. Efficient computation of permanents, with applications to boson sampling and random matrices. *Journal of Computational Physics*, 455:110990, 2022.
- [Mei21] Alex Meiburg. Inapproximability of positive semidefinite permanents and quantum state tomography. *arXiv preprint arXiv:2111.03142*, 2021.

- [MN62] Marvin Marcus and Morris Newman. Inequalities for the permanent function. *Annals of Mathematics*, pages 47–62, 1962.
- [Val79] Leslie G Valiant. The complexity of computing the permanent. *Theoretical computer science*, 8(2):189–201, 1979.
- [VY89] Vijay V Vazirani and Mihalis Yannakakis. Pfaffian orientations, 0–1 permanents, and even cycles in directed graphs. *Discrete Applied Mathematics*, 25(1-2):179–190, 1989.
- [Zec01] Riccardo Zecchina. Counting over non-planar graphs. *Physica A: Statistical Mechanics and its Applications*, 302(1-4):100–109, 2001.

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